

# SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

## Executive Summary

This report compares total site energy consumption, operational costs, construction costs, and payback period between baseline and all the applied renovation scenarios. The construction cost data is from 2026 RSMeans Database. Since RSMeans Database is proprietary, users can also adopt customized cost dataset when needed.

## Building Information

- Weather File: USA\_NY\_Buffalo-Greater.Buffalo.Intl.AP.725280\_TMY3.epw
- Building Name: SmallOffice
- Building Location: Buffalo, NY
- Total Floor Area: 511 m<sup>2</sup>

## Renovation Details by Scenario

| Scenario | Renovation Type | Renovation Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baseline | baseline        | <p><b>Current Condition of Exterior Wall:</b></p> <ul style="list-style-type: none"><li>• Exterior wall insulation: current R-value of insulation is R-11.3</li></ul> <p><b>Current Condition of Roof:</b></p> <ul style="list-style-type: none"><li>• Roof insulation: current R-value of insulation is R-1.2</li></ul> <p><b>Current Condition of Window:</b></p> <ul style="list-style-type: none"><li>• Number of window constructions is 20, with 20 simple glazing type windows, and 0 layered construction windows</li></ul> |

|            |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                             | <b>Current Condition of Door:</b> <ul style="list-style-type: none"> <li>Number of door constructions is 3, with 1 glass door, 0 overhead door, and 2 door</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Scenario 1 | wall + roof + window + door | <b>Wall upgrade:</b> <ul style="list-style-type: none"> <li>Exterior wall insulation: Blown Mineral Wool, target R-20.4</li> </ul> <b>Roof upgrade:</b> <ul style="list-style-type: none"> <li>Roof insulation: Blown Mineral Wool, target R-34.5</li> </ul> <b>Window upgrade:</b> <ul style="list-style-type: none"> <li>1-pane replacement</li> <li>Weatherstrip application (skipped: no OperableWindow)</li> <li>Glazing film application: safety film</li> <li>Caulking application: acrylic</li> <li>Window frame replacement: wood window frame</li> </ul> <b>Door upgrade:</b> <ul style="list-style-type: none"> <li>Bottom seal: brush weatherstrip</li> <li>Top/side seal: jamb weatherstrip</li> </ul> |
| Scenario 2 | wall + roof + window + door | <b>Wall upgrade:</b> <ul style="list-style-type: none"> <li>Exterior wall insulation: Blown Fiberglass, target R-20.4</li> </ul> <b>Roof upgrade:</b> <ul style="list-style-type: none"> <li>Roof insulation: Blown Fiberglass, target R-34.5</li> </ul> <b>Window upgrade:</b> <ul style="list-style-type: none"> <li>2-pane replacement</li> <li>Weatherstrip application (skipped: no OperableWindow)</li> <li>Glazing film application: anti-graffiti film</li> <li>Caulking application: polyurethane</li> <li>Window frame replacement: wood window frame</li> </ul> <b>Door upgrade:</b> <ul style="list-style-type: none"> <li>Bottom seal: silicone adhesive smoke gasket</li> </ul>                       |

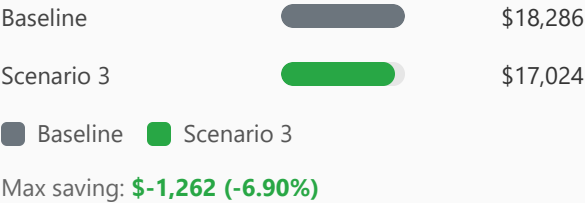
|            |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                             | <ul style="list-style-type: none"><li>Top/side seal: silicone adhesive smoke gasket</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Scenario 3 | wall + roof + window + door | <p><b>Wall upgrade:</b></p> <ul style="list-style-type: none"><li>Exterior wall insulation: Fiberglass Batts, target R-20.4</li></ul> <p><b>Roof upgrade:</b></p> <ul style="list-style-type: none"><li>Roof insulation: Fiberglass Batts, target R-34.5</li></ul> <p><b>Window upgrade:</b></p> <ul style="list-style-type: none"><li>2-pane replacement</li><li>Weatherstrip application (skipped: no OperableWindow)</li><li>Glazing film application: low-e film</li><li>Caulking application: polyurethane</li><li>Window frame replacement: wood-aluminium window frame</li></ul> <p><b>Door upgrade:</b></p> <ul style="list-style-type: none"><li>Door replacement: polystyrene core steel door (1 of 3 door(s) skipped: door type incompatibility)</li><li>Bottom seal: automatic door bottom</li><li>Top/side seal: silicone adhesive smoke gasket</li></ul> |

Annual Energy Analysis

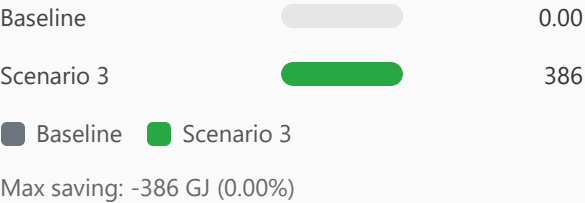
| Scenario   | Metric                       | Baseline | Renovation | Delta   | Savings % |
|------------|------------------------------|----------|------------|---------|-----------|
| Scenario 1 | Anual Total Site Energy (GJ) | 0.00     | 388.03     | -388.03 | 0.00%     |
| Scenario 2 | Anual Total Site Energy (GJ) | 0.00     | 388.03     | -388.03 | 0.00%     |
| Scenario 3 | Anual Total Site Energy (GJ) | 0.00     | 385.64     | -385.64 | 0.00%     |

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) (\$/yr) Comparison



Total Site Energy Saving (Scenario - Baseline) (GJ/yr) Comparison



Min Retrofit Construction Cost

\$30,731

Scenario 2

Lowest Retrofit Cost Payback Period

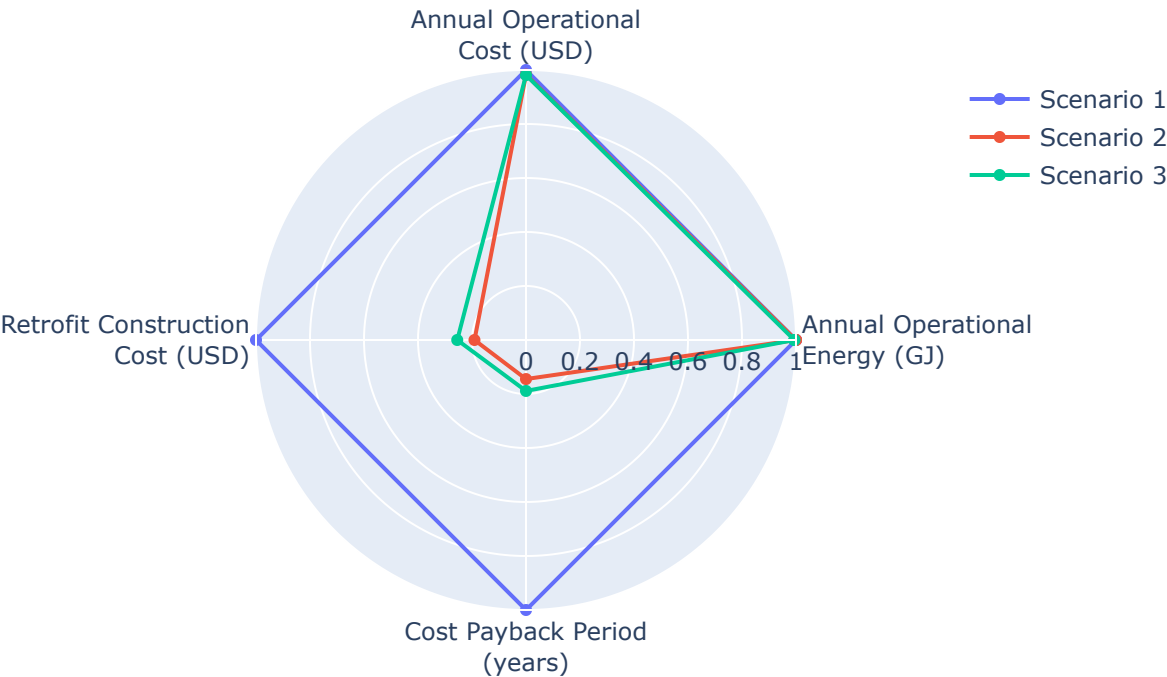
25 years

Scenario 2

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization

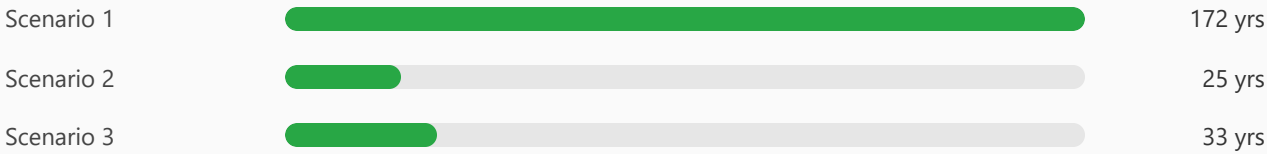
Parametric Scenario Comparison (Normalized Spider Chart)



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) (\$/yr)).



Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

| Scenario   | Component             | Material name                  | Volume (m³) | Area (m²) | Thickness (m) | Length (m) | Thermal Conductivity (W/m·K) | Density (kg/m³) | Product Lifetime (years) |
|------------|-----------------------|--------------------------------|-------------|-----------|---------------|------------|------------------------------|-----------------|--------------------------|
| Scenario 1 | Wall insulation       | Blown Mineral Wool             | 16          | 282       | N/A           | N/A        | 0.03                         | 90              | 30                       |
| Scenario 1 | Roof insulation       | Blown Mineral Wool             | 144         | 599       | 0.24          | N/A        | 0.03                         | 90              | 30                       |
| Scenario 1 | Door bottom sealing   | brush weatherstripping         | N/A         | N/A       | N/A           | 3.66       | N/A                          | N/A             | 15                       |
| Scenario 1 | Door top/side sealing | jamb weatherstripping          | N/A         | N/A       | N/A           | 16         | N/A                          | N/A             | 15                       |
| Scenario 2 | Wall insulation       | Blown Fiberglass               | 17          | 282       | N/A           | N/A        | 0.03                         | 30              | 30                       |
| Scenario 2 | Roof insulation       | Blown Fiberglass               | 144         | 599       | 0.24          | N/A        | 0.03                         | 30              | 30                       |
| Scenario 2 | Door bottom sealing   | silicone adhesive smoke gasket | N/A         | N/A       | N/A           | 3.66       | N/A                          | N/A             | 15                       |
| Scenario 2 | Door top/side sealing | silicone adhesive              | N/A         | N/A       | N/A           | 16         | N/A                          | N/A             | 15                       |

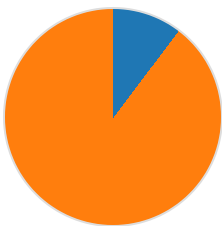
|            |                       |                                |     |     |      |      |      |     |    |
|------------|-----------------------|--------------------------------|-----|-----|------|------|------|-----|----|
|            |                       | smoke gasket                   |     |     |      |      |      |     |    |
| Scenario 3 | Wall insulation       | Fiberglass Batts               | 17  | 282 | N/A  | N/A  | 0.03 | 30  | 30 |
| Scenario 3 | Roof insulation       | Fiberglass Batts               | 144 | 599 | 0.24 | N/A  | 0.03 | 30  | 30 |
| Scenario 3 | Door bottom sealing   | automatic door bottom          | N/A | N/A | N/A  | 3.66 | N/A  | N/A | 15 |
| Scenario 3 | Door top/side sealing | silicone adhesive smoke gasket | N/A | N/A | N/A  | 16   | N/A  | N/A | 15 |

Result Summary

Per-scenario breakdown: left chart shows retrofit construction cost contribution by measure, right chart shows total site energy by fuel type (electricity, natural gas, and other). Water is excluded from energy breakdown.

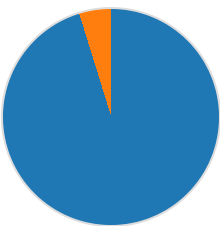
Scenario 1

Cost Breakdown by Retrofit Measure



■ Wall: \$2,664 (10.3%)  
■ Roof: \$23,226 (89.7%)  
Total: \$25,891

Annual Total Site Energy Breakdown (GJ)



■ Electricity: 369.41 GJ (95.2%)  
■ Natural Gas: 18.62 GJ (4.8%)  
Total: 388.03 GJ

Scenario 2

Cost Breakdown by Retrofit Measure



Annual Total Site Energy Breakdown (GJ)





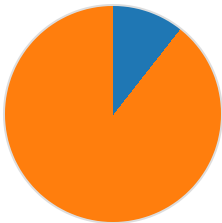
■ Wall: \$648 (10.6%)  
■ Roof: \$5,477 (89.4%)  
Total: \$6,125



■ Electricity: 369.41 GJ (95.2%)  
■ Natural Gas: 18.62 GJ (4.8%)  
Total: 388.03 GJ

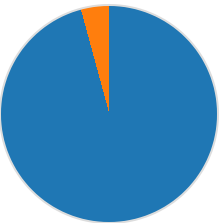
Scenario 3

Cost Breakdown by Retrofit Measure



■ Wall: \$1,742 (10.6%)  
■ Roof: \$14,727 (89.4%)  
Total: \$16,469

Annual Total Site Energy Breakdown (GJ)



■ Electricity: 369.34 GJ (95.8%)  
■ Natural Gas: 16.30 GJ (4.2%)  
Total: 385.64 GJ

| Scenario   | Retrofit Construction Cost (USD) | Annual Operational Cost (USD) | Anual Total Site Energy (GJ) |
|------------|----------------------------------|-------------------------------|------------------------------|
| Baseline   | \$0.00                           | \$18,286                      | 0.00                         |
| Scenario 1 | \$161,075                        | \$17,351                      | 388.03                       |
| Scenario 2 | \$30,731                         | \$17,052                      | 388.03                       |
| Scenario 3 | \$41,023                         | \$17,024                      | 385.64                       |